

Seeing Through Occlusion

Multi-Modal Motion Modeling for Video Instance Segmentation

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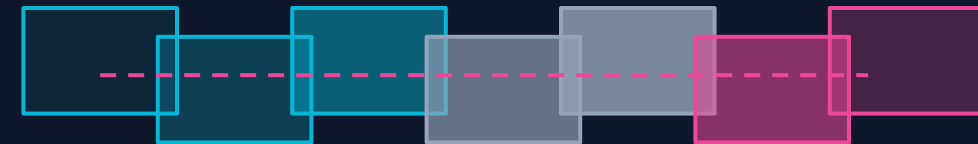
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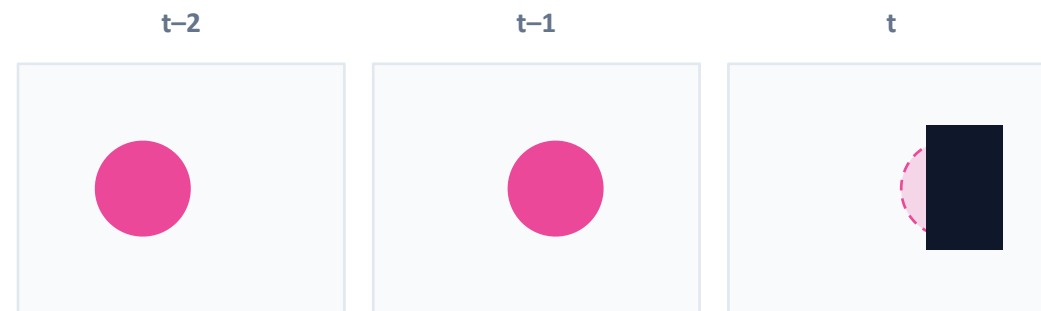


observed → occluded → predicted

Video Instance Segmentation collapses under occlusion

Standard Video Instance Segmentation (VIS) tracks objects by matching **appearance embeddings** between frames: CNN or transformer features that describe what the object looks like.

When objects are hidden behind obstacles, those embeddings **become unreliable or disappear entirely**. Identity flips, tracks break, and masks drift onto the wrong object.



Appearance says: "I don't see the object."

Motion says: "It's still moving along its trajectory."

20–30

mAP points

Performance gap on occlusion-heavy benchmarks

State-of-the-art VIS models lose 20–30 mAP when evaluated on the OVIS dataset compared to standard benchmarks.

OVIS

Occluded Video Instance Segmentation

901 videos

296K instance masks, 25 categories

5,223 uniquely-tracked instances

When you can't see it, track how it moves

Decouple tracking from appearance.

Model motion directly from bounding box trajectories.

1

Multi-modal motion

Decompose trajectory into velocity, acceleration, and shape-deformation streams

2

Occlusion-aware attention

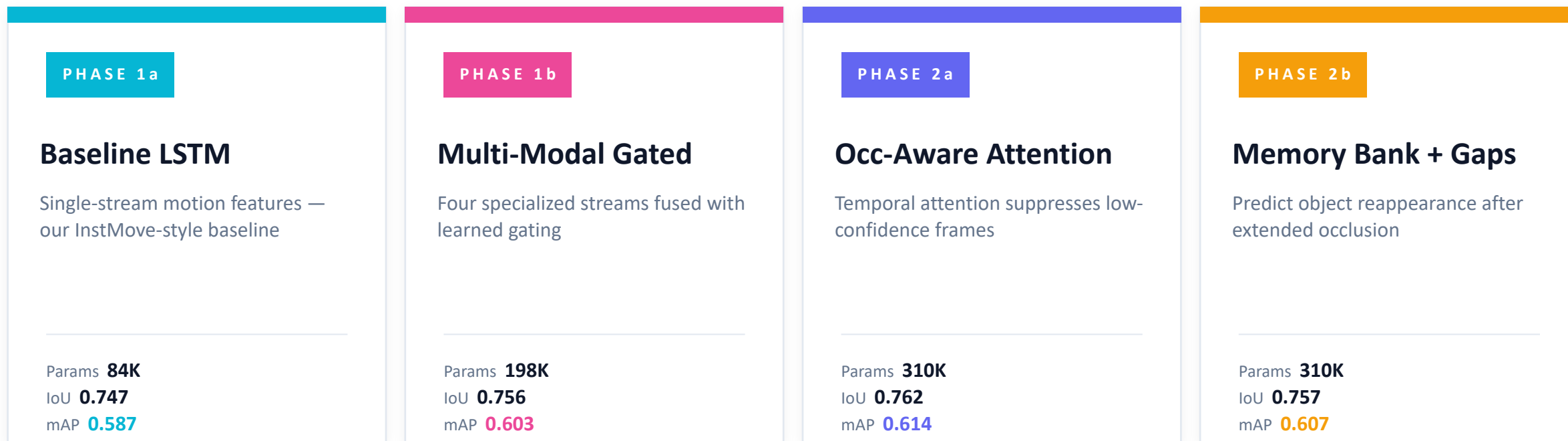
Suppress frames where occlusion confidence says the signal is noisy

3

Memory bank

Carry instance identity across long occlusion gaps

Four phases, each isolating one modeling decision

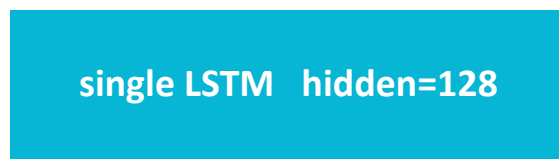
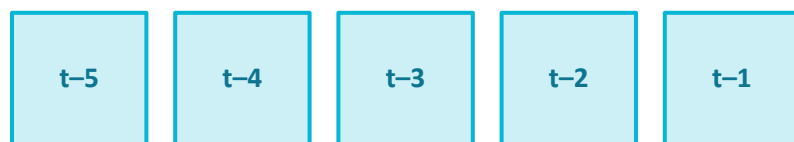


All numbers on OVIS validation set. Each phase builds directly on the previous checkpoint.



Single-stream motion LSTM: our InstMove baseline

10-D feature per frame



The 10-D feature vector

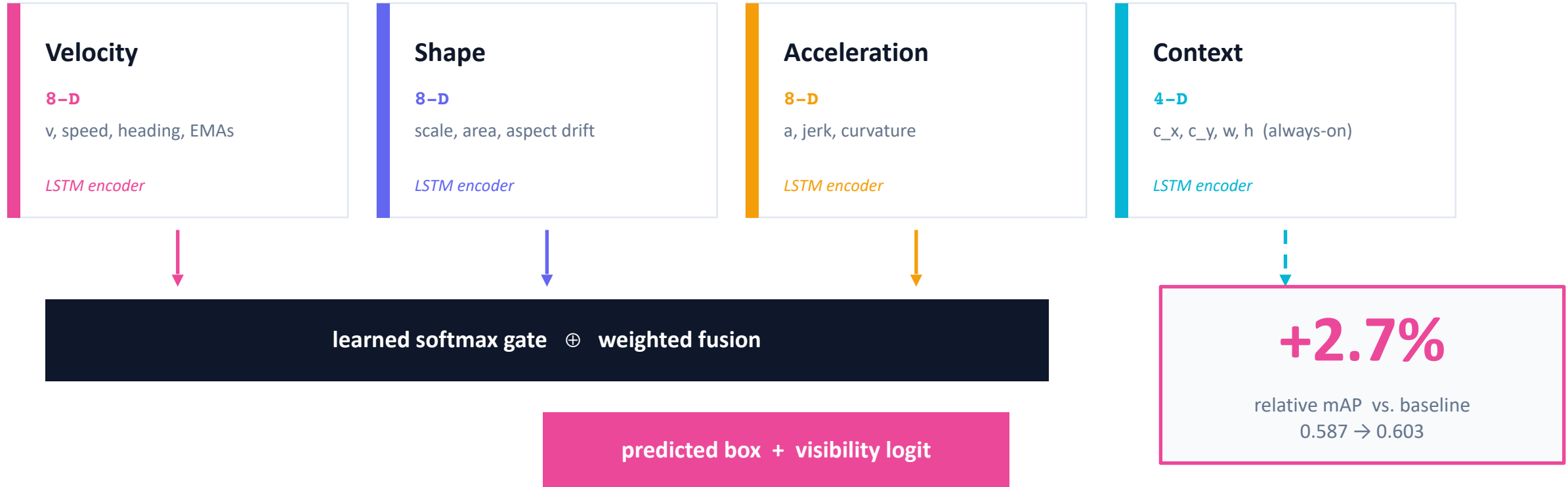
- Centroid c_x, c_y *position*
- Size w, h *box extent*
- Velocity v_x, v_y *first-order*
- Acceleration a_x, a_y *second-order*
- Area A *scalar*
- Log aspect $\log(w/h)$ *shape ratio*

84K
Params

0.747
Val IoU

0.587
Val mAP

Decompose motion into four specialized streams



Three design choices that mattered

Context not gated — position is always needed, never competitively weighted against motion.

Residual skip — predict box as last-observed + motion correction.

Light gate entropy reg — $\lambda = 0.01$ prevents collapse without forcing uniformity.

Occlusion-aware temporal attention

Three new modules on top of Phase 1b

1

Occlusion Confidence Module

Maps per-frame occlusion labels (none / slight / severe) + area ratios $\rightarrow$ scalar confidence in $[0, 1]$.

2

Occlusion-Aware Temporal Attention

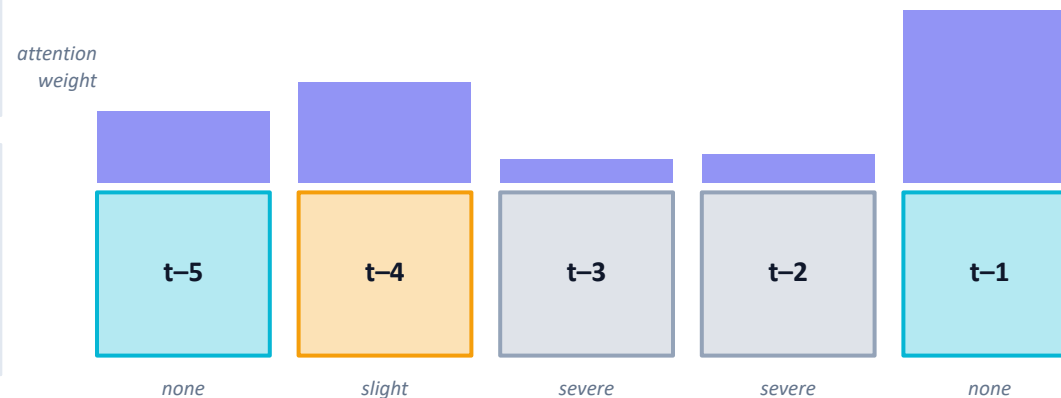
Scaled dot-product attention where the last timestep queries all history. Confidence added in log-space to pre-softmax scores — low-confidence frames are naturally downweighted.

3

Adaptive Gated Fusion

Extends Phase 1b gating by conditioning on a motion regime summary (mean speed, max acceleration, mean confidence).

What the attention learns



Occlusion $\Rightarrow$ lower attention. The model learns to lean on visible frames.

+1.8% mAP

over Phase 1b

Attention entropy

1.47 $\rightarrow$ 1.29

Memory bank for long-gap occlusion

The problem with temporal attention alone

Attention looks at recent history (5 frames). When an object is hidden for longer than that, the entire context window is unreliable — attention has nothing visible to attend to.

What the memory bank adds

- **Per-instance state** — Stores last visible box, velocity, confidence, frames since last seen
- **Gap-length embedding** — `nn.Embedding(50)` — learns distinct dynamics per gap length
- **Sigmoid read-out gate** — Balances memory representation vs. current noisy observation

The honest result. Gap training trades standard next-frame performance for gap-prediction ability. IoU drops 0.762 → 0.757. The best checkpoint for typical tracking remains Phase 2a. 2b matters when the object disappears for **>5 frames**.

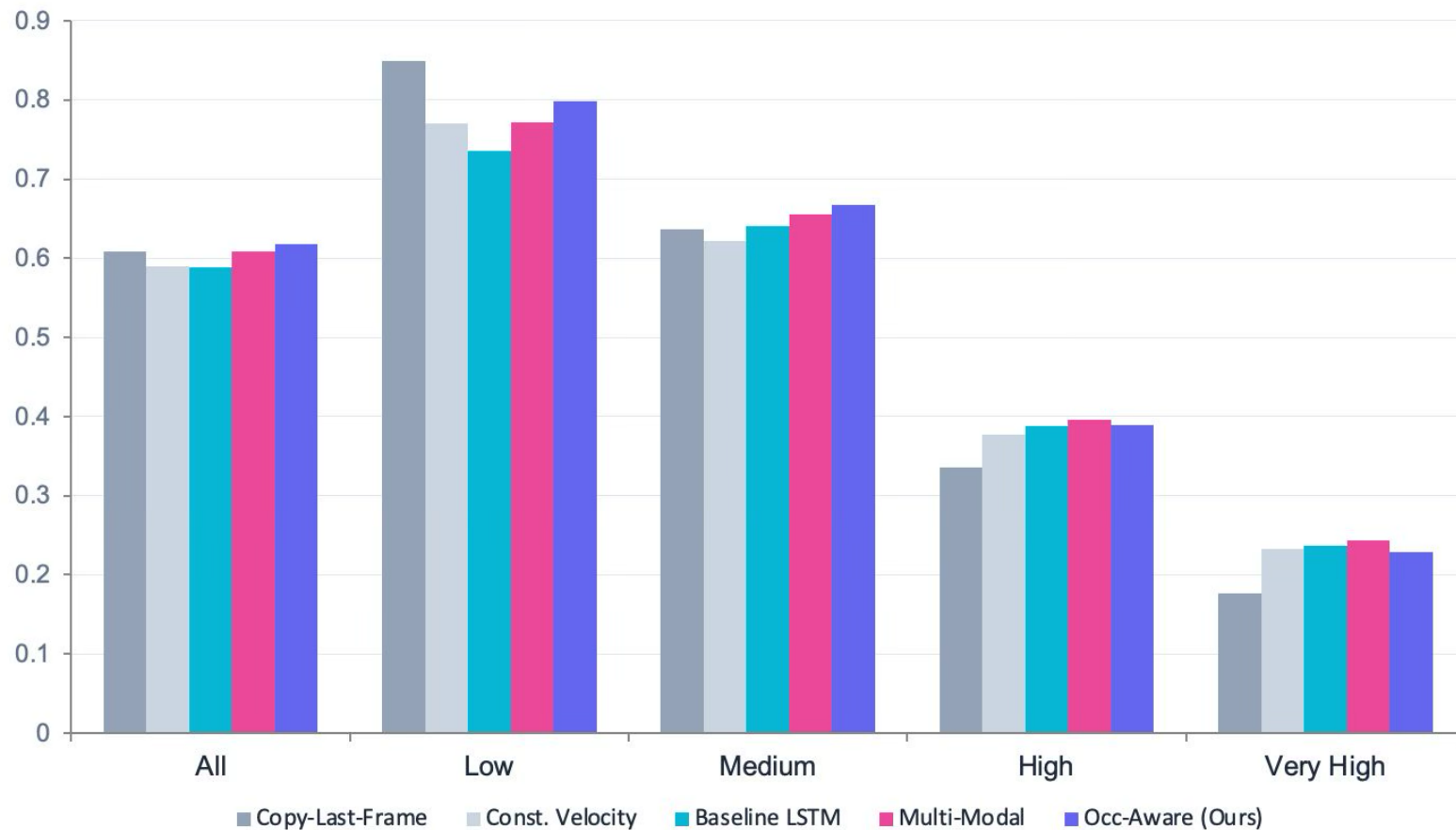
Why? Most samples are nearly-static. A model that does nothing wins on average. Overall mAP is a weighted average dominated by easy cases, so the hard ones, the ones where motion modeling actually matters, get drowned out.

Gap-length curriculum

<i>epochs 1–5</i>	gaps 1–3	n = 675
<i>epochs 6–10</i>	gaps 4–6	n = 169
<i>epochs 11+</i>	gaps 7–10	n = 118
	gaps 11+	n = 216

RESULTS

mAP by motion difficulty



Motion subsets partitioned by ground-truth centroid displacement: Low (bottom 33%), Medium (middle 33%), High (top 33%), Very High (top 10%).

Takeaways

0.618

best overall mAP

Occ-Aware tops every method

+17.8%

vs Copy-Last on High

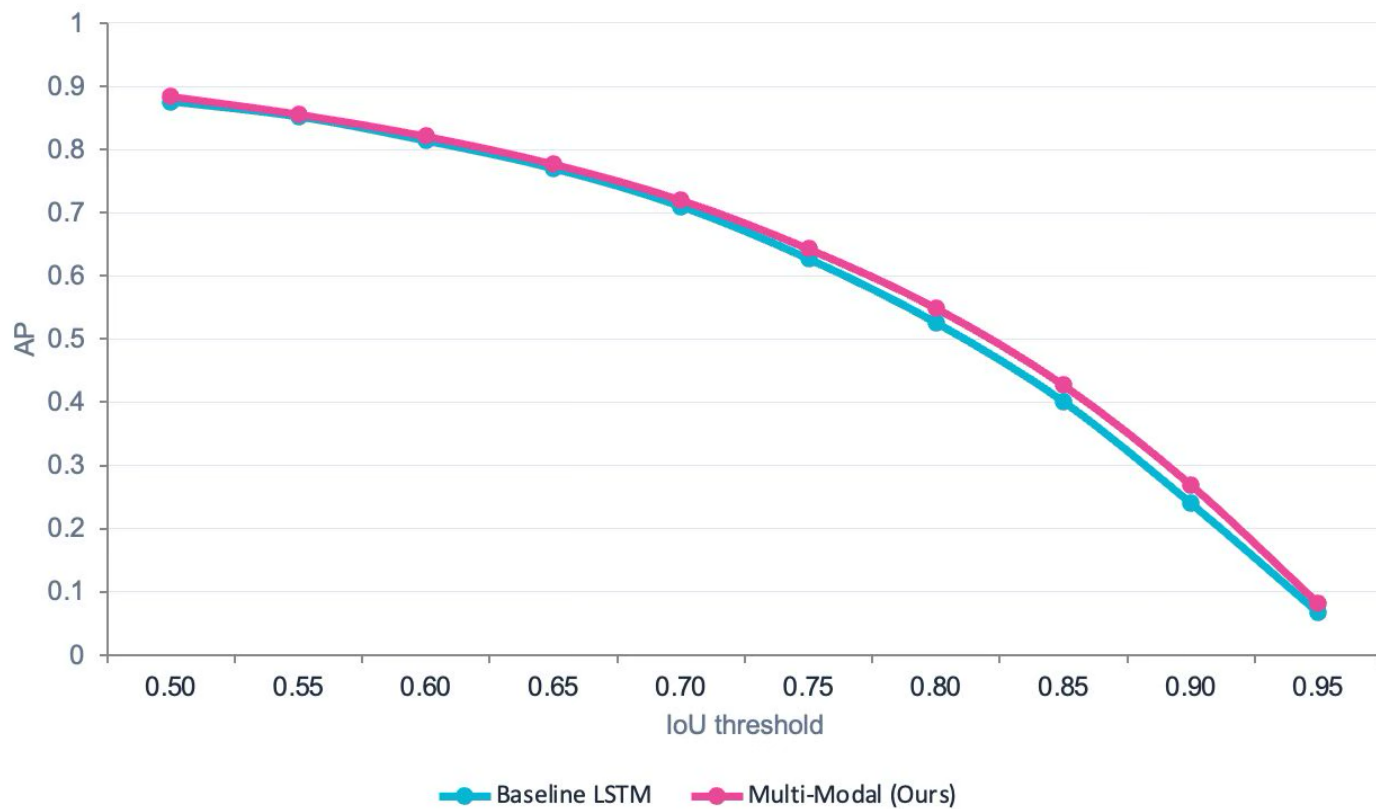
learned motion earns its keep

+38.0%

vs Copy-Last, Very High

gap scales with difficulty

Consistent gains at every IoU threshold



on high-motion samples (top 33% by displacement)

Where the gains concentrate

Improvements are largest at the strict thresholds ($AP_{0.80}-AP_{0.90}$), showing the multi-modal model produces tighter, better-localized boxes — not just vaguely correct ones.

ΔAP (Multi-Modal – Baseline)

AP @ 0.50	+0.008
AP @ 0.70	+0.010
AP @ 0.80	+0.023
AP @ 0.85	+0.026
AP @ 0.90	+0.029

SECTION 03

Live Demo

Running motion-based tracking on an OVIS clip in real time

WHAT TO WATCH FOR

- 1 **Tight tracking under partial occlusion** — Box stays locked even as the object passes behind obstacles
- 2 **Recovery after severe occlusion** — Trajectory picks up cleanly when the object reappears
- 3 **Baseline vs ours side-by-side** — Blue boxes are Phase 1a; pink are Phase 2a

Limitations & next steps

Limitations we still carry

- **Gate weights stay near uniform**
All three modality gates settle at ~ 0.33 . Motion-regime summary features aren't discriminative enough.
- **Bounding boxes only**
No mask-level shape features; PCA-of-mask proposal was deferred.
- **Small gap training set**
Only 1,178 visible-occluded-visible patterns — 118 in the 7–10 frame range.
- **2b destabilizes standard perf**
Gap objective and next-frame objective pull against each other during joint training.

What we'd do next

- **Per-instance adaptive gating**
Condition on raw feature magnitudes, not just summary statistics.
- **Mask-PCA shape stream**
Distinguish deformation (person bending) from perspective (object shrinking).
- **Synthetic gap augmentation**
Randomly mask visible frames in training to cheaply scale gap examples.
- **End-to-end with a VIS backbone**
Integrate with MaskTrack R-CNN or MinVIS for full OVIS benchmark evaluation.

TAKEAWAYS

Motion modeling pays off: but only where motion actually happens.

3-stream

gated fusion

velocity, shape, acceleration — decomposed motion
beats monolithic LSTM (+2.7% mAP)

attention

with occlusion priors

log-space confidence injection suppresses unreliable
frames (+1.8% mAP)

stratified

evaluation protocol

partitioning by displacement reveals the true value of
motion models

THANK YOU!